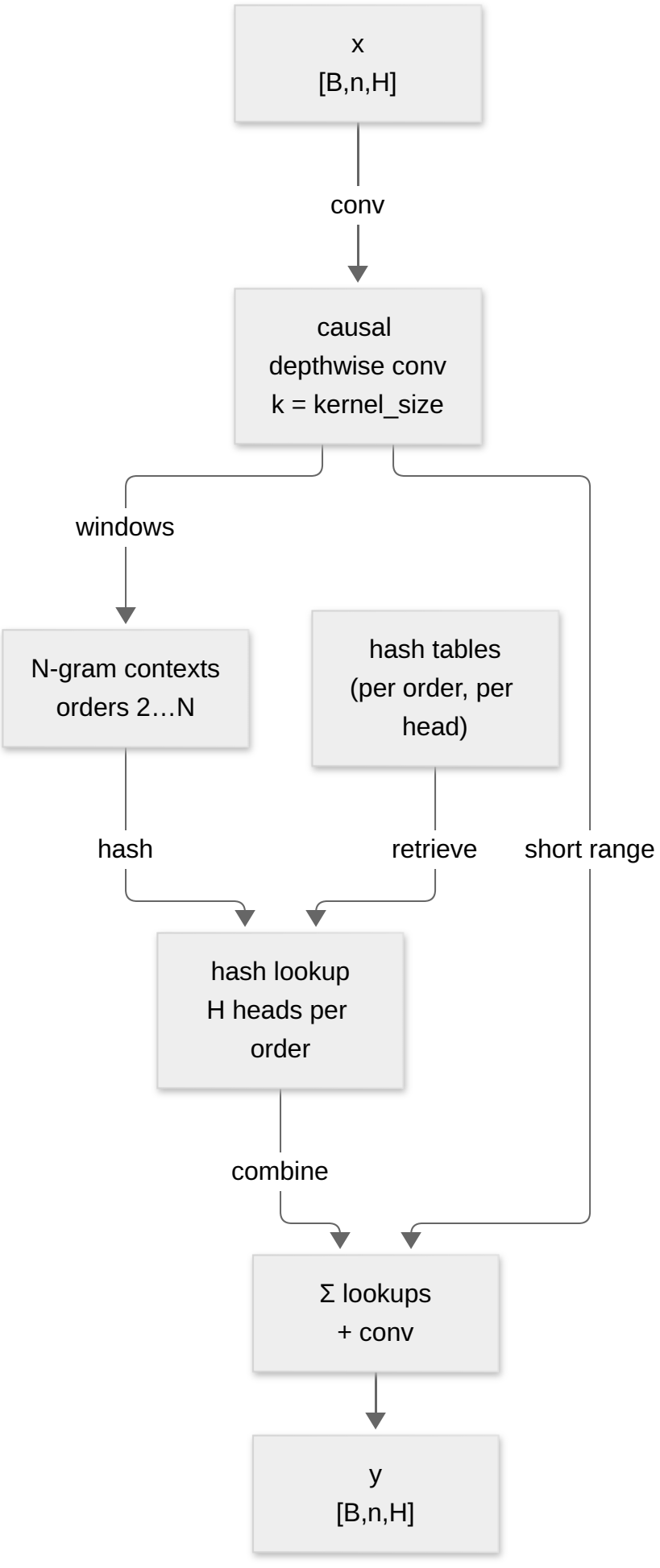

x
[B,n,H]
conv
causal
depthwise conv
k = kernel_size
windows
short range
N-gram contexts
orders 2…N
hash tables
(per order, per
head)
hash
retrieve
hash lookup
H heads per
order
combine
Σ lookups
+ conv
y
[B,n,H]